Temporal Graph Models for Bankruptcy Prediction: A Fiscal-Year Evaluation

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Abstract. Bankruptcy prediction is difficult because positive cases are rare. This study compares boosted-tree, recurrent, and graph models on a public SEC-derived financial dataset. After excluding 214 repeated company-year filings without reporting dates, the cohort contains 42,742 observations, 34 numeric features, and 293 next-year bankruptcy labels. Models are trained on 2008–2016, selected on 2017–2018, and evaluated on 2019–2021. XGBoost achieves test average precision (AP) 0.1141 and ROC-AUC 0.8868. The reliability-aware temporal GNN achieves AP 0.0383 and ROC-AUC 0.7515. XGBoost with GNN score fusion achieves AP 0.1133 and ROC-AUC 0.8806. A LightGBM–GraphSAGE blend reaches AP 0.1181 but has lower ROC-AUC than LightGBM; its paired AP confidence interval includes zero. A published XGBoost benchmark reports ROC-AUC 0.92 on a different cohort and evaluation protocol. Exact filing-availability dates are unavailable for this dataset, so the results use fiscal-year ordering and do not establish filing-time performance.

I. Introduction

Corporate bankruptcy is costly to firms, investors, employees, and lenders. Predictive models must find a small number of future events among many non-events, making average precision (AP) and recall at an operating point more informative than accuracy. Prior work on publicly traded U.S. firms reports that gradient-boosted trees are strong one-year-ahead predictors, establishing a demanding tabular baseline [1]. Graph neural networks (GNNs) have also been applied to bankruptcy and distress prediction, including models that combine corporate relations with temporal encoders [2], [3]. Thus, adding an LSTM to a GNN is not itself a novelty claim.

This study evaluates whether financial-peer graph messages improve bankruptcy ranking beyond firm-level temporal and boosted-tree models. The graph model encodes up to five years of company features with an LSTM and aggregates messages over relation-specific financial-similarity graphs. The evaluation includes tuned XGBoost and LightGBM baselines, conventional graph models, and validation-selected score fusion. Published results are reported separately when their datasets and evaluation protocols differ.

II. Related Work

Altman introduced a discriminant score for bankruptcy prediction [4]. Alanis et al. benchmarked eight methods on U.S. public firms using walk-forward one-year-ahead forecasts and report ROC-AUC 0.92 for XGBoost [1]. Table II lists this result separately because the cohort, predictors, and protocol differ from those used here. Zhao et al. combine intra-risk and contagion-risk signals with graph learning [2]. Sun et al. combine relation-specific graph convolution, relation attention, temporal encoding, and score fusion for distress prediction [3]. These studies establish temporal and relational graph learning as prior approaches to financial distress prediction.

The American Stock Market bankruptcy release initially considered for this project has a target constant within every company across its observed years, so it cannot support the intended next-year forecasting task as released [5]. We instead use the public SEC-derived numeric extension and its inherited next-year bankruptcy labels [6]. The thesis that introduced this extension reports stronger AP for its FinBERT text model than for its numeric-only variants; those published figures are context, not results from our run, and are not directly comparable with our numeric-only experiment.

III. Data and Task

A. Source and Cohort

The source contains filing-level rows joined by company identifier and SEC accession, a binary label, a year, and 34 dense numeric financial concepts. Its reported full cohort is 42,956 filings from 2008–2021. We audited the downloaded dense variant and found 214 rows belonging to repeated company-year groups (103 groups); one repeated group had conflicting labels. Since the release does not provide filing dates, selecting an amendment or duplicate as the canonical filing would require an unsupported temporal assumption. We therefore exclude every row in a repeated company-year group. The resulting cohort contains 42,742 observations, 5,612 firms, 293 positive observations (0.6855%), and 34 numeric predictors. Source rows and license files are not redistributed by this project; readers should download them from the repository record and cite the dataset author [6].

The source defines its inherited target as bankruptcy in the next fiscal year. Exact filing or publication dates are unavailable, and 103 repeated company-year groups cannot be ordered by fiscal year alone. The evaluation therefore uses fiscal-year splits and does not verify whether each feature was available on a specific prediction date.

B. Chronological Split

We train on fiscal years 2008–2016, select models and fusion weights on 2017–2018, and evaluate once on 2019–2021. The test set contains 9,032 rows and 64 positive

labels (0.7086% prevalence). Company identifiers occur in multiple years, so the primary split tests future years for a partially overlapping issuer population; the present report does not claim company-disjoint generalization. No test labels are used for feature transforms, model selection, or threshold selection.

IV. Methods

A. Preprocessing and Strong Tabular Baselines

Numeric features are converted to finite values, winsorized at the training-set 1st and 99th percentiles, median-imputed, and scaled by the training interquartile range. All transformation parameters are fitted on training years. We compare logistic regression, decision tree, random forest, histogram gradient boosting, MLP, LSTM, GCN, GraphSAGE, reliability-aware temporal GNN, LightGBM, and XGBoost. For each GBDT family we evaluate eight declared configurations on both the most recent-year input and flattened five-year history (16 validation trials), then select by validation AP. LightGBM selects 300 trees, learning rate 0.03, seven leaves, maximum depth four, 20 minimum child samples, and ℓ_2 penalty 1.0, with balanced class weighting. XGBoost selects depth two, learning rate 0.03, 300 trees, minimum child weight 10, ℓ_2 penalty 1.0, row and feature subsampling 0.8, and training-set positive-class weighting.

B. Reliability-Aware Temporal GNN

For firm i in year t, an LSTM maps a trailing sequence of up to five annual feature vectors to state $\mathbf{h}_{i,t}$. For each financial feature group r, we construct a directed k-nearest-neighbor graph ($k = 10$) using cosine similarity of the corresponding standardized features in year t. For edge (j, i), the base weight is $s_{ji,t} = \max(0.01, 1 - d_{\cos}(\mathbf{x}_{j,t}, \mathbf{x}_{i,t}))$. We down-weight unstable neighborhoods using the Jaccard overlap $p_{i,t}$ between firm i's peer sets in adjacent snapshots:

$$w_{ji,t}^{(r)} = s_{ji,t}^{(r)} \left(0.5 + 0.5 p_{i,t}^{(r)}\right). \tag{1}$$

Each relation channel computes a weighted mean of neighboring temporal states. A learned softmax gate combines relation messages, which are fused with the firm's own state and passed to a binary classifier. The GCN and GraphSAGE comparators use the same LSTM states and contemporaneous peer graph, with a single merged message channel. Financial similarity edges are predictive associations; they do not establish contagion or causal links.

C. Validation-Selected Score Fusion

We transform the tree and graph scores with robust location/scale parameters fitted on validation data, then evaluate convex combinations $q = (1 - \alpha) q_{\text{tree}} + \alpha q_{\text{graph}}$ for $\alpha \in \{0, 0.1, \ldots, 1\}$. We choose the combination with highest validation AP subject to validation ROC-AUC not falling below the tree score. The same validation-only rule selects $\alpha = 0.2$ for LightGBM–GraphSAGE, $\alpha = 0.1$ for LightGBM–reliability-GNN, and $\alpha = 0.1$ for XGBoost–reliability-GNN. XGBoost–GCN and XGBoost–GraphSAGE select $\alpha = 0$, so validation does not support adding either message-passing score under this rule.

D. Metrics and Uncertainty

AP is the primary metric and ROC-AUC is co-required. We also report Brier score for individual probability models and precision/recall at a threshold chosen on validation to target 80% recall. Fused standardized scores are rankings, not calibrated probabilities; we do not report Brier scores for them. Paired 95% confidence intervals compare test predictions by resampling companies with replacement and retaining all test-year rows for each sampled company (2,000 valid replicates). One training seed (42) and one future test window are available, so this interval does not capture seed or period variability.

V. Experimental Results

Table II preserves the published XGBoost benchmark value. It is context only, not a matched test-set result.

XGBoost is the strongest individual model in this evaluation on both AP and ROC-AUC, with AP 0.1141 and ROC-AUC 0.8868. The reliability-aware temporal GNN scores 0.0758 lower in AP and 0.1353 lower in ROC-AUC. GCN improves AP over the LSTM by 0.0086, while its ROC-AUC is lower. Each neural model uses one training seed.

The validation-selected LightGBM–GraphSAGE blend increases test AP from 0.1116 to 0.1181, while ROC-AUC decreases from 0.8790 to 0.8705. The paired company-clustered AP difference is 0.00515 with 95% interval [–0.00285, 0.01913]; the ROC-AUC difference is –0.00869 [–0.01783, –0.00065]. For XGBoost, validation selects a GNN weight of 0.1. The fused test AP is 0.1133 and ROC-AUC is 0.8806, compared with 0.1141 and 0.8868 for XGBoost alone. The paired AP-difference interval is [–0.01363, 0.01493], and the ROC-AUC difference is –0.00626 [–0.01558, –0.00037].

An additional diagnostic adds 12 conventional financial ratios to the 34 source features. Ratio-augmented LightGBM scores AP 0.0827 and ROC-AUC 0.8670. The ratio-augmented reliability-GNN scores AP 0.0286.

The originating dataset thesis reports AP 0.08 and ROC-AUC 0.87 for its raw numeric XBRL representation, and AP 0.30 and ROC-AUC 0.82 for its text FinBERT system [6]. These results use the unreduced filing cohort and are included as source-reported context.

VI. Conclusion

XGBoost achieved the highest individual AP and ROC-AUC among the evaluated models. The reliability-aware temporal GNN scored lower on both metrics, and its fusion with XGBoost also scored below the tree model alone. The LightGBM–GraphSAGE blend increased AP but decreased ROC-AUC, with an AP confidence interval

TABLE I
Held-out fiscal-year test results (2019–2021, $n = 9{,}032$, 64 positives). Operating thresholds target 80% recall on validation; final columns show test precision and recall.

Model	AP ↑	ROC-AUC ↑	Brier ↓	Precision (val 80%R)	Test recall
XGBoost, validation-tuned	0.1141	0.8868	0.0853	0.0262	0.8438
LightGBM, validation-tuned	0.1116	0.8790	0.0527	0.0239	0.8125
Logistic regression, last year	0.0504	0.8325	0.1693	0.0161	0.8125
Histogram gradient boosting, last year	0.0546	0.8028	0.0418	0.0204	0.8594
MLP, flattened five-year history	0.0589	0.8154	0.1846	0.0118	0.8281
LSTM	0.0708	0.7740	0.2067	0.0110	0.8750
GCN + LSTM	0.0794	0.7708	0.1701	0.0112	0.8281
GraphSAGE + LSTM	0.0698	0.7649	0.1782	0.0113	0.8750
Reliability-aware temporal GNN	0.0383	0.7515	0.2162	0.0120	0.7812
LightGBM + GraphSAGE score fusion	0.1181	0.8705	–	0.0252	0.8125
LightGBM + reliability-aware GNN fusion	0.1127	0.8688	–	0.0248	0.8125
XGBoost + reliability-aware GNN fusion	0.1133	0.8806	–	0.0256	0.8438

TABLE II
Published benchmark context from Alanis et al. The source reports ROC-AUC for XGBoost; AP is not reported. Different cohort, predictors, and walk-forward protocol make this non-comparable to our holdout.

Evaluation	AP	ROC-AUC
Alanis et al. [1]	n/r	0.9200
This study, XGBoost	0.1141	0.8868

that includes zero. Exact filing dates were unavailable, so these comparisons use fiscal-year ordering.

References

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